Experiment: 9 Date:29-7-26

Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim

To analyze the performance of a Massive MIMO system by studying channel capacity,
signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objectives

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR)
values using MATLAB. Observe how channel capacity increases with improving signal quality
based on the Shannon capacity principle.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB.
Analyze the effect of SNR on the quality and reliability of wireless communication in Massive
MIMO systems.

Objective – 3: Analyze Antenna Gain (dB)

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive
MIMO system using MATLAB. Observe how increasing the antenna array size improves
directional gain and enhances communication performance.

Software Required

• MATLAB

Theory

Massive MIMO employs a large number of antennas at the base station to simultaneously serve
multiple users. The increased spatial diversity and beamforming capability significantly improve channel capacity, signal quality, and antenna gain. Channel capacity is influenced by the signal-to-noise ratio (SNR), while antenna gain increases with the number of antenna elements. These parameters are fundamental to achieving the high throughput and spectral efficiency required in 5G and beyond wireless communication systems.

Objective – 1: Plot of Channel Capacity (bps/Hz) versus SNR (dB).

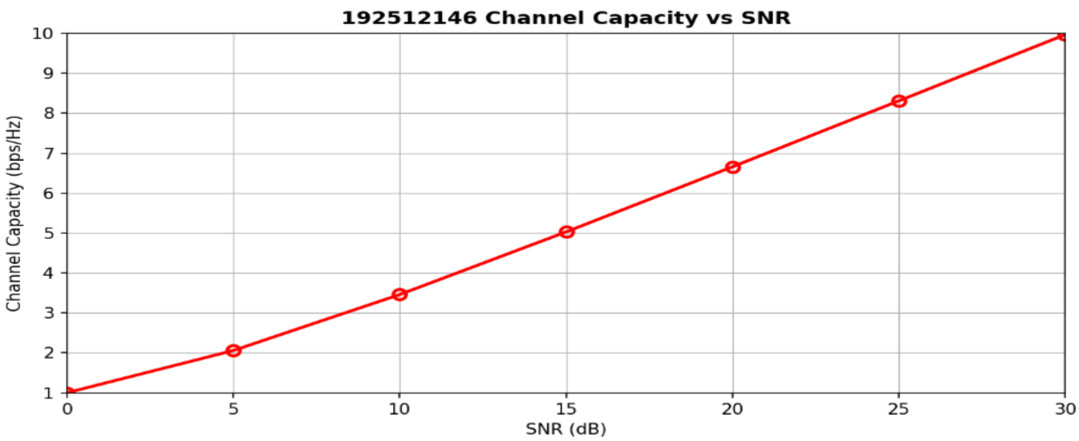
MATLAB Code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Expected Output

/MATLAB Drive/exp9.m

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6,'Color',"r");
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('192512146 Channel Capacity vs SNR');
```



Objective – 2 Bar graph comparing SNR values under different communication scenarios.
Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR)

values using MATLAB. Observe how channel capacity increases with improving signal quality
based on the Shannon capacity principle.

MATLAB Code:

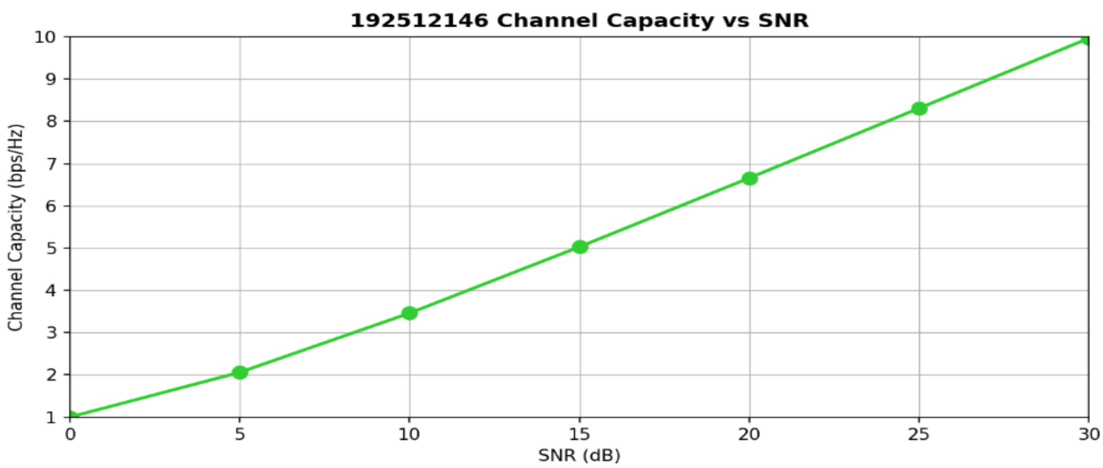
```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Expected Output

/MATLAB Drive/exp93.m

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6,'Color',"g");
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('192512146 Channel Capacity vs SNR');
```



Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

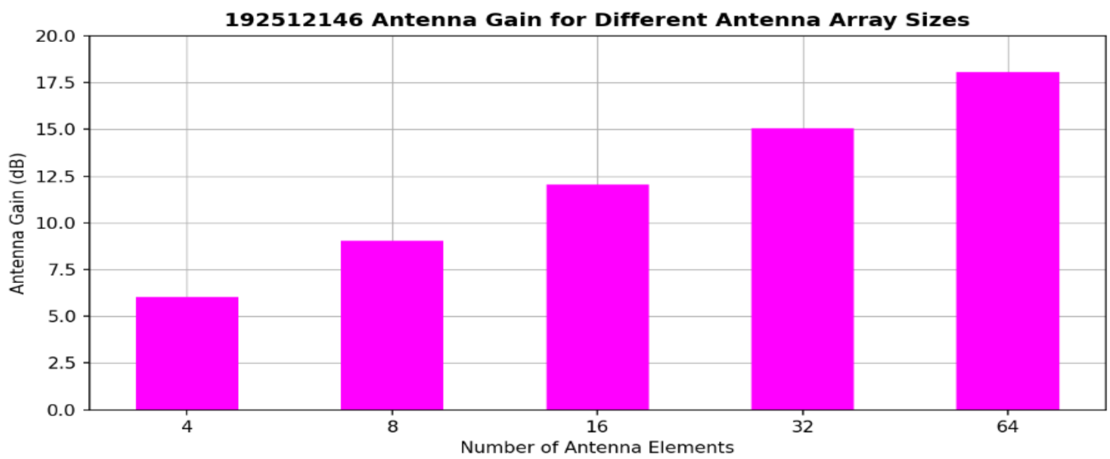
MATLAB Code:

```
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```

Expected Output

/MATLAB Drive/exp93.m

```
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain,"magenta")
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('192512146 Antenna Gain for Different Antenna Array Sizes')
```



Result

The MATLAB analysis successfully demonstrated the relationship between channel capacity,

signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher
SNR and larger antenna arrays improve communication performance by increasing channel
capacity and directional gain.